MASHARI ALREHAILI

Artificial Intelligence Student

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PROFESSIONAL SUMMARY

Artificial Intelligence student at Taibah University with a strong interest in AI automation and modern technologies. Focused on learning how intelligent systems are built and applied to real-world problems. Looking for opportunities to gain practical experience and further develop skills in the field of artificial intelligence.

EDUCATION

Bachelor – Artificial Intelligence | Taibah University 2022-09 – Present

PROJECTS

AI-Based Fire Detection System (System Design)

- Designed the architecture of a real-time AI-powered fire detection system capable of distinguishing fire from incense smoke, targeting 95%+ detection accuracy.
- Developed comprehensive system models including Use Case diagrams, Class diagrams, Sequence diagrams, and State diagrams using UML.
- Defined functional and non-functional requirements through scenario-based analysis, covering real-world retail environments.
- Proposed a layered system architecture (Presentation, Application, AI Engine, Data layers) with design patterns (Observer, Singleton, MVC).
- Planned a cross-platform user interface supporting live camera feed, alarm history, and configurable alert sensitivity for both mobile and desktop.

Network Intrusion Detection System Link

- Built a machine learning pipeline on the CIC-IDS 2017 dataset (2.83M records) to classify network traffic and detect cyberattacks including DDoS and port scanning.
- Preprocessed and cleaned raw data across 71 features, applying PCA to retain 18 components capturing ~89% of total variance.
- Trained and compared multiple models (XGBoost, Random Forest, CatBoost, LinearSVC); XGBoost achieved the best results with ~86% F1macro, 99%+ accuracy, and ROC-AUC of 0.9985.
- Addressed class imbalance via class-weighted training, boosting XGBoost F1-macro by over +10%.

Premier League Player Statistics Analysis (2023–2024) Link

- Analyzed a dataset of 581 players across 35 features, performing data cleaning including type correction, duplicate removal, and validation of negative values and invalid age ranges.
- Conducted statistical analysis to identify top scorers, assisters, and high-impact players using goals + assists combined metrics, incorporating xG and xAG for performance efficiency evaluation.
- Compared overperforming vs. underperforming players by measuring the difference between actual and expected goals (xG diff).
- Produced data visualizations covering team discipline, playing style (progressive metrics), goal efficiency (Actual vs. xG), nationality distribution, and age distribution using Python (pandas, matplotlib).

CERTIFICATIONS

Machine Learning and Deep Learning with Real-Time Demos – Saudi Data & AI Authority (SDAIA) 2026-01

Exploring Generative AI Use Cases – Saudi Data & AI Authority (SDAIA) 2026-01

Neural Networks and Deep Learning – DeepLearning.AI 2026-03

SKILLS

- **Python Programming**
- **Machine Learning**
- **Deep Learning**
- **Critical Thinking**
- **Problem Solving**

LANGUAGES

Arabic (Native) **English** (Professional Working)